ACADEMIC PROJECT

REPORT

Module: Deep Learning & Artificial Intelligence Engineering Cycle – 5th Year

BCMID
Breast Cancer Multimodal Imaging & Diagnosis

Design and Evaluation of an Intelligent Decision Support System for Breast Imaging Interpretation

Conducted by:
Ikbel Hamdi
Ahmed Fekih
Mohammed Malek Hammami
Samar Omrani

Supervised by:
Mme Oumayma Guesmi

Institution:

ESPRIT – École Supérieure Privée
d'Ingénierie et de Technologies

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Résumé

Le projet BCMID (Breast Cancer Multimodal Imaging & Diagnosis) vise à concevoir un prototype d’assistance à l’interprétation d’examens mammaires intégrant des données d’imagerie hétérogènes et des connaissances documentaires. L’approche relie la mammographie numérique plein champ (FFDM) et l’échographie à une architecture répartie en sept agents spécialisés, afin de produire un compte rendu structuré, explicable et soumis à vérification.

Le dispositif comprend un prétraitement des examens, deux branches ResNet-18, des cartes Grad-CAM, un système de recherche documentaire Clinical RAG (BAAI/bge-small-en-v1.5 et FAISS), un agent de synthèse piloté par Qwen2.5-3B-Instruct et une couche de sécurité déterministe. L’orchestration est assurée par n8n, en relation avec des microservices FastAPI.

Sur le corpus du projet (332 patientes, 2 001 images brutes, 321 rapports PDF), le sous-ensemble multimodal de 323 patientes obtient une ROC-AUC de 0,8198 (« malin contre autres »), contre 0,7917 pour la mammographie seule et 0,6743 pour l’échographie seule. L’audit XAI rapporte 87,4 % d’énergie Grad-CAM dans le masque mammaire ; 61/61 tests logiciels sont réussis. Ces résultats demeurent exploratoires : les annotations ne constituent pas une vérité terrain histologique, le corpus est limité et aucune validation clinique prospective n’est fournie.

Mots-clés : imagerie mammaire, fusion multimodale, ResNet-18, Grad-CAM, RAG, FAISS, Qwen2.5, agents intelligents, sécurité clinique.

Abstract

BCMID is an academic prototype supporting the interpretation of breast imaging by combining full-field digital mammography, ultrasound and clinical documentary knowledge through seven specialized agents (data quality, dual-branch ResNet-18 vision with Grad-CAM, FAISS-based clinical retrieval, Qwen2.5-3B-Instruct reasoning, deterministic safety checks and structured reporting), coordinated by n8n and FastAPI. On 323 patients with both modalities, multimodal ensemble ROC-AUC reaches 0.8198 versus 0.7917 (mammography) and 0.6743 (ultrasound); 61 software tests pass, 10/10 adversarial scenarios are neutralized and 17/20 reports are compliant on first pass. These observations do not establish medical effectiveness: labels derive from BI-RADS annotations, the cohort is small and no external clinical validation was conducted.

Keywords: multimodal breast imaging, explainable deep learning, clinical RAG, multi-agent orchestration, human oversight.

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Table of Contents

Résumé 2
Abstract 2
Table of Contents 3
General Introduction 4

Context and Multi-Agent Architecture 5
1.1 Clinical Context and Challenges 5
1.2 Problem Statement, Objectives and Scope 5
1.3 Justification of an Agentic Architecture 5
1.4 Global Architecture and the Seven Agents 6
Data and Multimodal Analysis 7
2.1 Medical Corpus and Annotation Distribution 7
2.2 Preprocessing, Quality Control and Leakage-Free Protocol 7
2.3 ResNet-18 Models and Multimodal Fusion 8
2.4 Explainability with Grad-CAM 8
Clinical RAG and Prompt Engineering 9
3.1 Knowledge Base and Semantic Search 9
3.2 Choice and Integration of Qwen2.5-3B-Instruct 9
3.3 Structuring Medical Prompts 9
Automation, Validation and Results 10
4.1 FastAPI Services and n8n Workflows 10
4.2 Control Loop and Automated Error Correction 10
4.3 Safety, Data Protection and Regulatory Boundaries 10
4.4 Multimodal Classification Performance 11
4.5 Agentic Pipeline Audit and Case Study 11
4.6 Critical Discussion and Threats to Validity 12
General Conclusion 12

Bibliographical References 12

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General Introduction

Breast cancer represents a major public health concern, for which the quality of medical imaging interpretation plays a decisive role. Breast examination mobilizes mammography, ultrasound, clinical records and standardized diagnostic reference systems: these sources are highly complementary, yet their heterogeneity demands a rigorous methodology to minimize interpretation errors.

Deep learning enables the automated extraction of rich representations useful for classification, and explainable AI (XAI) techniques such as Grad-CAM visualize regions associated with model decisions. Nonetheless, an elevated probability score does not constitute clinical proof, and an activation heatmap does not confirm the anatomical presence of a lesion. Large language models (LLMs) facilitate the generation of structured summaries, yet a fluent response may contain numerical discrepancies or ungrounded assertions. It is therefore vital to segregate image analysis, documentary knowledge retrieval and report synthesis, while subjecting all outputs to deterministic, independent checks.

BCMID combines two ResNet-18 vision branches with Grad-CAM, a Clinical RAG retrieval module (FAISS) and a Qwen2.5-3B-Instruct model, structured across seven specialized agents orchestrated via n8n and FastAPI. The central goal is to study how an explainable, traceable multimodal architecture can articulate visual predictions, documentary groundings and report generation while strictly preserving human supervision. BCMID is an academic research and demonstration prototype, and not an approved clinical medical device.

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CHAPTER 1

Context and Multi-Agent Architecture

1.1 Clinical Context and Challenges

Full-Field Digital Mammography (FFDM, CC and MLO views) enables the detection of microcalcifications and subtle architectural distortions; ultrasound characterizes the solid versus cystic nature of an anomaly, its margins and its posterior acoustic behavior. Combining modalities aims to exploit non-identical physical information. Interpretation becomes significantly more challenging in dense breast tissue, causing sensitivity to vary [1].

Figure 1.1: Examples of mammography and ultrasound data from the project image dataset.

1.2 Problem Statement, Objectives and Scope

How can multiple medical modalities be analyzed jointly, their results linked to documentary evidence, and a structured summary generated whose coherence can be verified prior to specialist review? Four core challenges emerge: D1 Data Heterogeneity; D2 Interpretability (revealing influential areas without framing them as confirmed lesions); D3 Documentary Grounding (tying syntheses to identifiable citations); D4 Output Safety (preventing contradictory, inaccurate or unreviewed statements).

Scope: BCMID is an academic decision-support prototype, not an approved medical device nor an autonomous diagnostic system. A BI-RADS category derived from a numeric score matrix must never be confused with a clinical radiological assessment performed according to the ACR Atlas [2]. The human_review_required field materializes this essential safeguard.

1.3 Justification of an Agentic Architecture

BCMID does not delegate all operations to a single generative model. Images are preprocessed and verified before analysis, probabilities originate from dedicated vision encoders, references are retrieved via RAG, and LLM text is audited prior to dispatch. Dividing the pipeline into autonomous agents segregates ingestion, vision prediction, knowledge retrieval, report writing and safety auditing; n8n sequencing renders inputs, outputs and rejection points traceable at every phase. This justification is architectural: it does not imply an empirical benchmark across alternative orchestrators.

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1.4 Global Architecture and the Seven Agents

Figure 1.2: BCMID Architecture: seven agents, n8n orchestrator, infrastructure, monitoring and clinical end-users.

Agent Responsibilities and Methods Technologies Outputs

01 – Ingestion Reception, metadata verification, SHA-256 hashing,

pseudonymization Python, pandas Verified dossier,

pseudonymized ID

02 – Data Quality Profiling, Otsu thresholding, OCR, inpainting, patient-level

splitting OpenCV, EasyOCR Normalized tensors,

quality report

03 – Vision Dual ResNet-18, multimodal fusion, scores, Grad-CAM

attention PyTorch, torchvision Probabilities, uncertainty,

heatmaps

04 – Clinical RAG Semantic encoding, dual FAISS indices, cross-fold patient
filtering

bge-small-en-v1.5, FAISS
IndexFlatIP

Grounded excerpts [G1],
[E1]

05 – Reasoning Evidence aggregation, schema-constrained structured

synthesis Qwen2.5-3B-Instruct Structured JSON report

proposition

06 – Quality &
Safety

Numerical tolerance, semantic grounding, citation
validation Deterministic rules, pytest Approval, revision or

flagged alert

07 – Reporting Delivery in structured multi-target formats (doctor /

technician)

ReportLab, FastAPI,
Pydantic, n8n

PDF report, JSON,
interactive Web view

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CHAPTER 2

Data and Multimodal Analysis

2.1 Medical Corpus and Annotation Distribution
The patient level constitutes the primary grouping unit in order to preserve absolute independence between training and validation sets.

Dataset Metric Volume Specification / Description
Registered Patients 332 327 of which feature supervised labels
Raw Medical Images 2,001 Before quality control and artifact filtering
Retained Images 1,974 951 mammograms and 1,023 ultrasound scans
Medical PDF Reports 321 320 retained for clinical RAG after filtering
Bimodal Patients 323 Complete cohort for ROC-AUC benchmarking
Overall Volume 12.11 GB Documented across raw archive packages

Lesion categories comprise normal (15.1%), benign (64.5%), malignant (19.0%) and five unlabelled BI-RADS 0 cases (1.5%). The benign class predominance constrains aggregated metric interpretation. The supervised target is derived from BI-RADS annotations rather than independent histological proof: the evaluation measures alignment with project annotations.

Figure 2.1: Breakdown of the 332 patients across annotation categories.

2.2 Preprocessing, Quality Control and Leakage-Free Protocol
• Profiling and Integrity: Formats, salted pseudonymized IDs, duplicates, corrupted scans, metadata concordance.
• Spatial Preprocessing: Breast region segmentation via Otsu thresholding, background masking, 224×224 resize, normalization.
• Burn-in Text Removal: Ultrasound metadata detected via EasyOCR, masked and inpainted using Navier–Stokes algorithms.
• Patient Partitioning: 5-fold cross-validation with GroupKFold; Clinical RAG strictly excludes current patient and fold documents.

Figure 2.2: Examples of preprocessing, polarity correction and burn-in text inpainting.

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Methodological Precaution: An image-level split is insufficient when a patient possesses multiple scans. Patient-level grouping and RAG document isolation represent two distinct, mandatory safeguards against data leakage, to be audited upon every reproduction cycle.

2.3 ResNet-18 Models and Multimodal Fusion

Each imaging modality is fed into a single-channel 224×224 ResNet-18 encoder [3], accounting for distinct textures and physics. Four fusion strategies are evaluated: late fusion (score averaging), weighted logit fusion, uncertainty-guided fusion (weight inversely proportional to Shannon entropy) and multimodal ensemble combining previous models with test-time augmentation (TTA). Shannon entropy H(p) = -Σ p

c

log2

p

c

serves as an internal uncertainty metric; it does not turn
outputs into clinically calibrated probabilities. The benchmark covers 323 patients (GroupKFold, malignant vs other) without undocumented hyperparameter tuning.

2.4 Explainability with Grad-CAM

Grad-CAM [4] utilizes target class gradients with respect to convolutional feature maps: α

k

c

= (1/Z) Σ

i,j ∂y

c

/∂Ai,j

k
and L

c
=

ReLU(Σ

k

α
k

c
Ak

). The spatial audit evaluates 387 heatmaps across 64 validation dossiers: average activation energy inside
the segmented breast parenchyma reaches 87.4% (with 12.6% outside). This confirms spatially plausible attention, without claiming systematic lesion correspondence.

Figure 2.3: Original scans, activation heatmaps and Grad-CAM overlays for mammography and ultrasound.

Articulation between Vision and Documentary Agents: The vision agent transmits raw logits, probabilities, uncertainties and Grad-CAM visual overlays; documentary and generative agents must never modify these numerical values. High ROC-AUC and plausible heatmaps do not replace histological benchmarks or radiologist sign-off: hence the mandate for independent deterministic safety checks.

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CHAPTER 3

Clinical RAG and Prompt Engineering

3.1 Knowledge Base and Semantic Search

Clinical RAG retrieves documentary passages describing rules and relevant observations so that the generative LLM justifies its phrasing against an explicit grounding context; it documents the synthesis without certifying clinical veracity. Texts are extracted, cleaned, chunked, encoded via BAAI/bge-small-en-v1.5 (384 dimensions) and indexed in FAISS IndexFlatIP (inner product of unit vectors, equivalent to cosine similarity [5, 6]). Two separate vector stores are retained:
• Guidelines Index: Nine prototype rules derived from ACR BI-RADS principles (non-exhaustive, non-official);
• Cases Index: 320 sanitized historical patient reports (one excluded), chunked into 995 passages.
Token lengths and overlap are not documented as clinically validated. Query search returns up to four passages ([G1] or [E1]), whose validity is confirmed downstream. Two filters mitigate data leakage: text_blind (redacting BI-RADS categories, cancer confirmations or surgeries) and exclude_pseudo_id (removing current patient and fold records).

3.2 Choice and Integration of Qwen2.5-3B-Instruct

The selection of Qwen2.5-3B-Instruct [7] is motivated by its compact footprint, local on-premise execution and superior instruction-following capability (occupying ~3.1 GB VRAM on a Tesla T4 GPU in project testing). The LLM was deliberately not fine-tuned: domain adaptation relies entirely on structured prompt engineering and RAG context, while supervised fine-tuning was appropriately applied to the vision backbones. Greedy decoding at temperature=0.0 minimizes stochastic drift without asserting absolute determinism. No comparative LLM benchmark is documented.

3.3 Structuring Medical Prompts

Prompt Section Integrated Information Targeted Constraint

SYSTEM Synthesis persona, clinical neutrality, strict ban on fabricating

observations

Prevent ungrounded claims, enforce JSON
grammar

CONTEXT Vision logits, uncertainty, Grad-CAM summary, authorized history,
RAG passages

Segregate calculated facts from documentary
citations

TASK Cross-modal reconciliation, interpretation and proposed follow-up Ensure medical coherence with provided

context

OUTPUT Mandatory schema keys and expected data types Enable automated deterministic JSON

validation

The output schema enforces six keys: predicted_birads, confidence, interpretation, recommendations, limitations, human_review_required. The confidence metric represents an internal heuristic, not a validated clinical probability. A total of 190 prompts were constructed (64 image-only, 62 text-only, 64 multimodal); these figures reflect experimental setup and do not claim statistical superiority across conditions.

Evaluation of RAG: No quantitative table of precision, Recall@4 or chunking comparisons was provided in the baseline source; no synthetic metric is fabricated. Nine rules and 995 chunks do not constitute a full validation of retrieval quality. Prompt iteration logs are maintained separately from experimental laboratory traces.

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CHAPTER 4

Automation, Validation and Results

4.1 FastAPI Services and n8n Workflows
Processing steps are exposed as FastAPI REST endpoints; n8n orchestrates sequential execution, enforces routing conditions and maintains an audit trail. Two workflows are documented: BCMID-Medical-Analysis (8-node main pipeline, webhook to structured report) and Triage (9 nodes: ingestion, ID extraction, report retrieval, conditional branching, dispatch). Triage relies on the logical flag requires_immediate_escalation; urgent branches trigger on heuristic BI-RADS 4A–5 categories. These flows simulate hospital IT and do not constitute an active production deployment inside a certified PACS/HIS.

4.2 Control Loop and Automated Error Correction
The Reasoning ↔ Quality & Safety coupling is essential: LLM proposals undergo deterministic validation independent of generated assertions. Upon detecting an anomaly, Agent 6 refuses approval and issues targeted corrective feedback; two revision cycles are authorized before flagging the dossier as FLAGGED_FOR_MANUAL_CLINICIAN_REVIEW.

Audit Check Deterministic Verification Rule
Numerical Consistency Tolerated gap between quoted and vision probability: |∆p| ≤ 0.006
Uncertainty Guard When prototype entropy H ≥ 0.50, mandate explicit cautionary phrasing
Clinical Entities Reject medical findings not attested by vision models or authorized dossier text
Citation Integrity Every reference token must match a valid chunk in current retrieval context
Supervision Enforcer Enforce explicit disclaimer; unconditionally impose human_review_required = true

4.3 Safety, Data Protection and Regulatory Boundaries
A continuous malignancy score is mapped to an indicative category and internal workflow priority. This table is purely heuristic and not clinically validated; it must neither be confused with ACR BI-RADS diagnostic criteria nor guide real patient treatment without a radiologist.

Score pmal Indicative Category Internal Priority Action
p < 0.02 BI-RADS 1 Routine screening interval
0.02 ≤ p < 0.10 BI-RADS 2 Routine screening interval
0.10 ≤ p < 0.35 BI-RADS 3 Short-interval 6-month surveillance
0.35 ≤ p < 0.65 BI-RADS 4A Workflow escalation (Urgent notification)
0.65 ≤ p < 0.85 BI-RADS 4B Workflow escalation (Urgent notification)
0.85 ≤ p < 0.95 BI-RADS 4C Workflow escalation (Urgent notification)
p ≥ 0.95 BI-RADS 5 Workflow escalation (Urgent notification)

The architecture implements patient ID pseudonymization via salted SHA-256 and burn-in ultrasound text inpainting; on-premise execution of FAISS and the LLM limits cloud exfiltration. Pseudonymization is not permanent anonymization: key management, access logs and retention policies remain mandatory. GDPR, HIPAA and European MDR 2017/745 [8] form regulatory frameworks to address; the prototype does not claim formal certification.

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4.4 Multimodal Classification Performance
Reported metrics address distinct objectives: classification (323 patients), XAI audit (387 heatmaps) and software resilience tests, each using specific samples and criteria.

Model Configuration ROC-AUC Clinical Sensitivity
Mammography Alone (ResNet-18) 0.7917 76.2 %
Ultrasound Alone (ResNet-18) 0.6743 68.4 %
Late Fusion (Averaging) 0.8150 81.5 %
Weighted Logit Fusion 0.8160 81.8 %
Uncertainty-Guided Fusion 0.8120 80.9 %
Multimodal Ensemble with TTA 0.8198 83.1 %

Figure 4.1: Comparison of reported ROC-AUC scores across configurations (axis truncated at 0.60).

The ensemble outperforms standalone mammography by +0.0281 ROC-AUC and ultrasound by +0.1455; sensitivity increases from 76.2% to 83.1%. Variations between fusion methods remain modest and no formal significance test or confidence interval was documented in baseline logs.

4.5 Agentic Pipeline Audit and Case Study
Audited Target Outcome Scope and Interpretation
Constructed Prompts 190 64 image-only; 62 text-only; 64 multimodal
Adversarial Tests 10 / 10 Prompt injections and conflicting constraints neutralized
Compliant on 1st Pass 17 / 20 3 reports corrected after first revision cycle (85% initial pass)
n8n Executions 23 5 initial execution failures (21.7%) documented and resolved
Mean Runtime 5.54 s Standard n8n logger indicator across recorded runs
Full Successful Run 12.984 s Observed end-to-end execution including Grad-CAM and PDF
pytest Test Suite 61 / 61 100% passed in 1.84 s within test environment

Case Study: Pseudonymized patient Paacb1ec57b (bilateral mammography and targeted ultrasound) yields a prototype

category of BI-RADS 3, pmal = 0.12, reported confidence of 78%, Shannon entropy H = 0.38 and 89.2% Grad-CAM activation energy inside the parenchymal mask. This illustrates a nominal output instance, requiring formal specialist sign-off.

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4.6 Critical Discussion and Threats to Validity

Structural Limitations: • Cohort: 332 patients, class imbalance, 323 for multimodal comparison. • Ground Truth: Labels derived from BI-RADS annotations without independent histological validation. • Generalization: No multicentric external validation nor prospective trial. • Comparisons: No confidence intervals, statistical significance tests between fusions, nor LLM cross-benchmarking. • XAI: 12.6% of Grad-CAM attention energy resides outside segmented breast tissue. • Integration: 5 n8n failures out of 23; 3 out of 20 reports required active revision. • Clinical Decision: Score-to-BI-RADS mapping and entropy gating thresholds lack clinical validation.

General Conclusion

The project integrates complementary scientific contributions: joint analysis of two medical imaging modalities, comparative evaluation of fusion strategies, XAI attention maps, a local dense vector documentary base and deterministic LLM output gating by independent rules. The system's core strength lies not in any isolated algorithm, but in the separation of responsibilities between vision prediction, knowledge retrieval, textual generation and deterministic safety verification.

Multimodal fusion achieves a higher ROC-AUC than either standalone modality (0.8198), Grad-CAM provides spatial visual evidence, RAG provides transparent access to normative excerpts, and reasoning/safety agents structure and verify the resulting clinical report.

Perspectives: External multi-hospital validation with independent pathology records; stability audit of n8n production workflows; quantitative benchmarking of RAG retrieval (Recall@4, citation accuracy); score calibration; and clinical review of triage rules prior to any clinical deployment. This contribution remains that of an academic research prototype, in which every output is traceable, all uncertainty disclosed, and all interpretation subject to human expertise.

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